Burali-Forti paradox: there is no set of all the ordinals

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the transitivity derivation, the promotion to an ordinal, and the closing contradiction all appear with their three named internal citations verbatim in the span.

Editorial note. The label buraliforti names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Elements of ordinals are ordinals. Same unit, \ollabel{ordmemberord} (source line 416). Line 555: transitivity of the supposed set O of all ordinals, since members of its members are ordinals.
- Transitive sets of ordinals are ordinals. Same unit, \ollabel{corordtransitiveord} (source line 534). Line 556: O itself becomes an ordinal once transitive.
- No ordinal contains itself. Same unit, \ollabel{ordordered} (source line 517). Line 557: O in O contradicts the irreflexivity supplied by the ordinal-ordering corollary.

Source reference. Original corollary corordtransitiveord: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 534–534. Prerequisite pointer invoked at proof line(s) 557; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ordmemberord: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 416–416. Prerequisite pointer invoked at proof line(s) 555; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ordordered: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 517–517. Prerequisite pointer invoked at proof line(s) 558; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Burali-Forti Paradox)**.** *There is no set of all the ordinals*

Proof. For reductio, suppose O is the set of all ordinals. If $\alpha \in \beta \in O$, then α is an ordinal, by ordmemberord, so $\alpha \in O$. So O is transitive, and hence O is an ordinal by corordtransitiveord. Hence $O \in O$, contradicting ordordered. □